Supplementary Material for TriGAC

A Complete Time Complexity Analysis

Let n be the number of objects, m the number of views, k the number of ground-truth clusters, and T the number of interaction rounds. Let M_t denote the number of constraint-graph WCCs after round t, and Q_{t-1} the number of queries recorded before round t. The representative-window size is bounded by Ω. We analyze algorithmic runtime, excluding human response time.

Object Graph Initialization. In each view, **TriGAC** uses a ball-tree for nearest-neighbor searches. Under efficient pruning, tree construction and nearest-neighbor searches take expected log-linear time in the number of nodes at each level. Since each node has one outgoing edge before representative-agent selection, WCC extraction and representative-agent selection take linear time. At each nonterminal level, every WCC contains at least two nodes. Selecting one representative agent per WCC therefore at least halves the number of nodes in the next higher-level graph. Consequently, the total expected time per view is

$$\mathcal{O}\Big(n\log n+\frac{n}{2}\log\frac{n}{2}+\frac{n}{4}\log\frac{n}{4}+\cdots\Big)=\mathcal{O}(n\log n).$$

Computing the static representativeness scores from the initialized graph takes an additional $\mathcal{O}(n)$ time per view. Thus, initialization across all views takes expected $\mathcal{O}(mn\log n)$ time. This expected bound relies on efficient ball-tree pruning; without effective pruning, exact nearest-neighbor searches can require quadratic time per view.

View Graph Construction and View-Confidence Update. The first round initializes uniform view weights in $\mathcal{O}(m)$ time and skips view-graph construction. In each later round, computing NMI for all pairs of view-specific partitions takes $\mathcal{O}(m^2n)$ time. Using these precomputed similarities, hierarchical merging over the views takes $\mathcal{O}(m^2)$ time. The resulting sparse view graph contains $\mathcal{O}(m)$ edges, so sorting its edge weights, applying Kneedle, pruning edges, and extracting WCCs together take $\mathcal{O}(m\log m)$ time. View-graph construction therefore takes $\mathcal{O}(m^2n)$ time per round.

For confidence estimation, we compute the total prediction sum over all views and the prediction sum within each view cluster for every recorded query. These sums require $\mathcal{O}(mQ_{t-1})$ time and allow leave-one-cluster-out and leave-one-view-out predictions to be obtained by subtraction, rather than independently recomputing each view combination. Since there are at most m view clusters and m views, evaluating the prediction errors and marginal contributions at both levels takes $\mathcal{O}(mQ_{t-1})$ time. Normalizing the resulting confidence weights takes $\mathcal{O}(m)$ time. Let Q be the total number of recorded queries. Summing over the rounds gives

$$\mathcal{O}\left(\sum_{t=2}^{T}\big(m^2n+mQ_{t-1}\big)\right)\subseteq\mathcal{O}(Tm^2n+TmQ).$$

Under noiseless feedback, each processed object generates at most k queries. Since objects are processed without replacement, $Q\le nk$. Thus, this module takes $\mathcal{O}(Tm^2n+Tmnk)$ time in total.

Informative Object Selection. In the first round, the per-view representativeness scores are already available from initialization. Normalizing these scores and computing $\mathcal{R}_1(x_i)$ for all objects take $\mathcal{O}(mn)$ time. Sorting the objects by $\mathcal{R}_1$ takes $\mathcal{O}(n\log n)$ time. Checking whether the final representative agent of each view is in the selected batch and adding any missing agents takes $\mathcal{O}(m)$ time using an indexed membership array.

For each later round, let $p_t=|\mathcal{P}_t|\le n$ be the number of unselected candidates. Each candidate is compared with at most Ω objects in each of the M_{t-1} representative windows in every view. Predicting its nearest WCC in each view therefore takes $\mathcal{O}(mM_{t-1}\Omega)$ time. The assignment cost for all candidates is $\mathcal{O}(p_tmM_{t-1}\Omega)$. Computing confidence-weighted uncertainty, normalizing the static representativeness

scores, and aggregating them across views take $\mathcal{O}(p_t m)$ time. Ranking candidates by uncertainty and breaking ties by aggregated representativeness take $\mathcal{O}(p_t \log p_t)$ time.

Under noiseless feedback, a new constraint-graph WCC is created only when the processed object receives CL outcomes against all existing WCCs. Hence, each WCC represents a distinct ground-truth cluster and $M_t \leq k$. Using $p_t \leq n$ and $\Omega \geq 1$, the total cost of object selection is therefore

$$\mathcal{O}\left(mn + n\log n + \sum_{t=2}^{T}\big(p_t m M_{t-1}\Omega + p_t m + p_t \log p_t\big)\right) \subseteq \mathcal{O}(mn\log n + Tnmk\Omega + Tn\log n).$$

Constraint Graph Update. For each selected object, at most k existing WCCs are considered under noiseless feedback. Computing distances in all views to the union of representative windows for each WCC takes $\mathcal{O}(m^2 k\Omega)$ time. Normalizing and aggregating these distances take $\mathcal{O}(m^2 k\Omega)$ time, while sorting the WCCs takes $\mathcal{O}(k\log k)$ time. Sequential queries stop at the first ML outcome or after all existing WCCs return CL. Recording at most k query outcomes and their m per-view predictions takes $\mathcal{O}(mk)$ time.

Each processed object either joins one existing WCC or creates a singleton WCC. Its membership and anchor information can therefore be maintained incrementally without recomputing all constraint-graph WCCs. Because the representativeness scores are static, updating the affected representative window requires only inserting the new object into the retained top-Ω objects, taking $\mathcal{O}(m\Omega)$ time across all views. Thus, the computational cost per processed object is

$$\mathcal{O}\big(m^2 k\Omega + mk + k\log k + m\Omega\big) = \mathcal{O}\big(k(m^2\Omega + \log k)\big).$$

At most n objects are processed across all rounds, so this module takes $\mathcal{O}\big(nk(m^2\Omega + \log k)\big)$ time.

Object Graph Refinement and Consensus Partition Fusion. In round t, each processed object has at most one outgoing edge removed and at most one ML edge added in each view. With indexed outgoing-edge storage, these local updates take $\mathcal{O}(m|\mathcal{S}_t|)$ time. The refined object graphs retain at most one outgoing edge per object and therefore contain $\mathcal{O}(n)$ edges each. Extracting their WCCs and updating the view-specific partitions take $\mathcal{O}(mn)$ time per round.

For final consensus fusion, each object receives one label from each view. Accumulating confidence-weighted votes only for these predicted labels takes $\mathcal{O}(m)$ time per object, rather than scanning all possible labels. The final fusion therefore takes $\mathcal{O}(mn)$ time. Since objects are selected without replacement, $\sum_{t=1}^{T}|\mathcal{S}_t| \leq n$. The total cost of refinement, partition extraction, and final fusion is consequently

$$\mathcal{O}\left(m\sum_{t=1}^{T}|\mathcal{S}_t| + Tmn + mn\right) = \mathcal{O}(Tmn).$$

Overall Time Complexity. Combining the costs of the five modules gives

$$\begin{aligned}\mathcal{T} \in \mathcal{O}\big(&mn\log n + Tm^2 n + TmQ\\ &+ Tnmk\Omega + Tn\log n\\ &+ nk(m^2\Omega + \log k) + Tmn\big).\end{aligned}$$

Using $Q \leq nk$, $\Omega \geq 1$, $T \geq 1$, and $1 \leq k \leq n$, we obtain the overall expected bound

$$\mathcal{T} \in \mathcal{O}\big(mn\log n + Tnk(m^2\Omega + \log n)\big) \tag{1}$$

The dataset-dependent quantities m and k are typically constant with respect to n, while T and Ω are user-specified constants in **TriGAC**. As a result, **TriGAC** has expected $\mathcal{O}(n\log n)$ time complexity.

B Theoretical Analysis

1) Unique-Cycle Property

Theorem B.1 (Unique-Cycle Property)**.** *In the maximum co-cluster probability graph, each weakly connected component contains exactly one directed cycle.*

Proof. We prove the theorem by contradiction, structured in two parts.

Part 1: Existence of at least one directed cycle in a WCC.

Assume for contradiction that there exists a weakly connected component (WCC) C that contains no directed cycles. Let $n_C \triangleq |C|$ denote the number of objects in C. Since every object in C has exactly one outgoing edge to its maximum co-cluster probability neighbor, we can construct a directed path sequence starting from any object $x_i \in C$:

$$x_i \to \psi(x_i) \to \psi^2(x_i) \to \psi^3(x_i) \to \cdots$$

where $\psi(x_i) \triangleq \arg\max_{x_k \neq x_i} p(x_i, x_k)$ denotes the maximum co-cluster probability neighbor of x_i, and $\psi^t(x_i)$ denotes the object reached by following t outgoing edges starting from x_i:

$$\psi^t(x_i) \triangleq \underbrace{\psi(\psi(\cdots\psi}_{t \text{ times}}(x_i)\cdots)).$$

Under the assumption that C contains no directed cycles, all objects in the sequence must be distinct.

(I) When $t = 1$, the untraversed object set is $C \setminus \{x_i, \psi(x_i)\}$, the number of traversed objects is 2, and the number of traversed edges is 1.

(II) When $t = n_C - 1$, all n_C objects in C have been traversed. Hence, the untraversed object set is empty, and $n_C - 1$ edges have been traversed.

(III) When $t = n_C$, since $\psi^{n_C-1}(x_i) \in C$, it must have exactly one outgoing edge. However, the set of untraversed objects is empty. Therefore, $\psi^{n_C}(x_i)$ must be one of the objects already traversed. Let $\psi^{n_C}(x_i) = \psi^j(x_i)$ for some $0 \leq j \leq n_C - 1$. This forms a directed cycle:

$$\psi^j(x_i) \to \psi^{j+1}(x_i) \to \cdots \to \psi^{n_C-1}(x_i) \to \psi^{n_C}(x_i) = \psi^j(x_i)$$

which contradicts the initial assumption that C contains no directed cycles. Therefore, every weakly connected component must contain at least one directed cycle.

Part 2: Uniqueness of the directed cycle in a WCC.

Assume for contradiction that there are two distinct directed cycles in C, denoted by $\mathcal{L}_1$ and $\mathcal{L}_2$. Let $V(\mathcal{L}_1)$ and $V(\mathcal{L}_2)$ be the sets of objects in cycles $\mathcal{L}_1$ and $\mathcal{L}_2$, respectively. We consider two cases based on whether these cycles share objects.

Case 1: $V(\mathcal{L}_1) \cap V(\mathcal{L}_2) = \emptyset$.

Since every object $x_i \in C$ has exactly one outgoing edge, the traversal sequence starting from any object is unique. As shown in Part 1, this path must eventually enter a directed cycle and remain within it. For any object $x \in V(\mathcal{L}_1)$, the unique path starting from x is entirely contained within $\mathcal{L}_1$, meaning $\psi^k(x) \in V(\mathcal{L}_1)$ for all $k \geq 0$. Similarly, for any object $y \in V(\mathcal{L}_2)$, the unique path starting from y is entirely contained within $\mathcal{L}_2$, meaning $\psi^k(y) \in V(\mathcal{L}_2)$ for all $k \geq 0$. Now, consider any object $z \in C$. The path starting from z must eventually enter some directed cycle. If it enters $\mathcal{L}_1$, then all subsequent objects $\psi^k(z)$ (for sufficiently large k) belong to $V(\mathcal{L}_1)$. If it enters $\mathcal{L}_2$, they belong to $V(\mathcal{L}_2)$. Since $V(\mathcal{L}_1)$ and $V(\mathcal{L}_2)$ are disjoint, a path cannot lead to both cycles. Moreover, for every directed edge $z \to \psi(z)$, z and $\psi(z)$ eventually reach the same cycle. Hence no edge in the underlying undirected graph connects objects that reach different cycles. The two cycles therefore cannot be connected by an undirected path, contradicting that both lie in the same WCC C.

Case 2: $\mathcal{L}_1$ and $\mathcal{L}_2$ share at least one object.

Let z be an object such that $z \in V(\mathcal{L}_1)$ and $z \in V(\mathcal{L}_2)$. Since z is in cycle $\mathcal{L}_1$, repeatedly applying the function ψ starting from z must trace the objects of $\mathcal{L}_1$ in sequence and eventually return to z. Let this sequence be:

$$z \to \psi(z) \to \cdots \to \psi^{l_1}(z) = z.$$

Similarly, since z is in cycle $\mathcal{L}_2$, repeatedly applying the function ψ starting from z must trace the objects of $\mathcal{L}_2$ in sequence and eventually return to z. Let this sequence be:

$$z \to \psi(z) \to \cdots \to \psi^{l_2}(z) = z.$$

Since each object has exactly one outgoing edge, the successor of z, which is $\psi(z)$, is unique. The successor of $\psi(z)$, which is $\psi^2(z)$, is also unique, and so on. Therefore, the sequence generated by starting at z and repeatedly applying ψ is entirely determined and unique. This unique sequence must form the directed cycle containing z. Since this sequence defines both $\mathcal{L}_1$ and $\mathcal{L}_2$, it must be that $\mathcal{L}_1$ and $\mathcal{L}_2$ are the exact same directed cycle. This contradicts the assumption that $\mathcal{L}_1$ and $\mathcal{L}_2$ were distinct directed cycles.

Both cases lead to a contradiction. Therefore, the assumption that a weakly connected component C contains more than one directed cycle is false. Combining the results from both parts, we conclude that every weakly connected component C contains exactly one directed cycle. □

2) Cycle Compactness Property

Lemma B.2 (Distance Monotonicity)**.** *Given a set of objects $\{x_1, \ldots, x_n\}$, if $x_b = \psi(x_a)$ and $x_c = \psi(x_b)$, then*

$$d(x_a, x_b) \geq d(x_b, x_c). \tag{2}$$

Proof. Since $x_c = \psi(x_b)$ is the maximum co-cluster probability neighbor of x_b, and co-cluster probability decreases monotonically with distance, x_c is also the nearest neighbor of x_b in terms of distance. Therefore:

$$d(x_b, x_c) = \min_{x \neq x_b} d(x_b, x).$$

This implies that there does not exist any object x such that $d(x_b, x) < d(x_b, x_c)$; otherwise, x_c would not be the maximum co-cluster probability neighbor of x_b. Consequently, we have:

$$d(x_a, x_b) \geq d(x_b, x_c).$$

□

Corollary B.3 (Monotonicity along Outgoing Paths)**.** *Given a WCC C in the maximum co-cluster probability graph, for any node $x_i \in C$, consider the path formed by repeatedly following outgoing edges:*

$$x_i \to \psi(x_i) \to \psi^2(x_i) \to \psi^3(x_i) \to \cdots.$$

Then the outgoing edge lengths along this path are non-increasing:

$$d\big(x_i, \psi(x_i)\big) \geq d\big(\psi(x_i), \psi^2(x_i)\big) \geq d\big(\psi^2(x_i), \psi^3(x_i)\big) \geq \cdots. \tag{3}$$

Proof. For any $\tau \geq 0$, consider two consecutive outgoing edges

$$\psi^\tau(x_i) \to \psi^{\tau+1}(x_i), \quad \psi^{\tau+1}(x_i) \to \psi^{\tau+2}(x_i).$$

Applying Lemma B.2 with $x_a = \psi^\tau(x_i)$, $x_b = \psi^{\tau+1}(x_i)$, and $x_c = \psi^{\tau+2}(x_i)$ yields

$$d\big(\psi^\tau(x_i), \psi^{\tau+1}(x_i)\big) \geq d\big(\psi^{\tau+1}(x_i), \psi^{\tau+2}(x_i)\big).$$

Chaining this inequality over $\tau = 0, 1, 2, \ldots$ establishes the result. □

Theorem B.4 (Cycle Compactness Property)**.** *Let C be a WCC of the maximum co-cluster probability graph in view v, let $\mathcal{L}$ be its unique directed cycle, and let $\psi(\cdot)$ denote the outgoing neighbor with the maximum co-cluster probability. Then:*

(i) ***Uniformity.*** *For any two cycle nodes $u, w \in \mathcal{L}$,*

$$p(u, \psi(u)) = p(w, \psi(w)). \tag{4}$$

(ii) ***Maximality.*** *For any cycle node $u \in \mathcal{L}$ and any off-cycle node $x \in C \setminus \mathcal{L}$,*

$$p(u, \psi(u)) \geq p(x, \psi(x)). \tag{5}$$

Proof. Part (i). Consider any node $u_t \in \mathcal{L}$ and index the cycle nodes as

$$u_1 \to u_2 \to \cdots \to u_{n_\mathcal{L}} \to u_1,$$

where $u_{t+1} = \psi(u_t)$ for $t = 1, \ldots, n_\mathcal{L}$ (with $u_{n_\mathcal{L}+1} = u_1$). Applying Corollary B.3 to the consecutive edges $u_t \to u_{t+1}$ and $u_{t+1} \to u_{t+2}$ yields

$$d(u_t, u_{t+1}) \geq d(u_{t+1}, u_{t+2}).$$

Chaining this inequality over all nodes on the cycle, we obtain the circular sequence

$$d(u_1, u_2) \geq \cdots \geq d(u_{n_\mathcal{L}-1}, u_{n_\mathcal{L}}) \geq d(u_{n_\mathcal{L}}, u_1) \geq d(u_1, u_2).$$

Since the leftmost and rightmost terms are identical, all intermediate inequalities must hold with equality. Therefore,

$$d(u_1, u_2) = \cdots = d(u_{n_\mathcal{L}}, u_1) \triangleq d_\mathcal{L}.$$

Because co-cluster probability decreases monotonically with distance, we have

$$p(u, \psi(u)) = p(w, \psi(w)), \quad \forall u, w \in \mathcal{L}. \tag{6}$$

Part (ii). For any off-cycle node $x \in C \setminus \mathcal{L}$, let its absorption path be

$$x \to \psi(x) \to \psi^2(x) \to \cdots \to \psi^{t-1}(x) \to u,$$

where $u = \psi^t(x) \in \mathcal{L}$ is the first node on the path that belongs to $\mathcal{L}$. By Corollary B.3, the edge lengths along this path are non-increasing:

$$d(x, \psi(x)) \geq \cdots \geq d(\psi^{t-1}(x), u) \geq d(u, \psi(u)).$$

Since $u \in \mathcal{L}$, Part (i) implies $d(u, \psi(u)) = d_{\mathcal{L}}$. Thus, by transitivity,

$$d(x, \psi(x)) \geq d_{\mathcal{L}}.$$

Again, because co-cluster probability decreases monotonically with distance, we have

$$p(u, \psi(u)) \geq p(x, \psi(x)), \quad \forall u \in \mathcal{L},\ x \in C \setminus \mathcal{L} \tag{7}$$

□

3) Local Density at the Cycle Scale

Theorem B.4 shows that the cycle edges have equal length and are no longer than off-cycle outgoing edges within the same WCC. This gives a local-density separation at the cycle-edge cutoff when pairwise distances have no ties.

Definition B.5 (Local Density [1])**.** *For a cutoff distance $d_c > 0$, the local density of x_i within a WCC C is*

$$\rho_i(d_c) = \sum_{\substack{x_j \in C \\ j \neq i}} \mathbf{1}_{\{d(x_i, x_j) \leq d_c\}}. \tag{8}$$

Corollary B.6 (Local density at the cycle scale)**.** *Let C be a WCC with unique directed cycle $\mathcal{L}$, and let $d_{\mathcal{L}}$ be the common length of its cycle edges. Suppose distinct unordered pairs of objects in C have distinct distances. At the cutoff $d_c = d_{\mathcal{L}}$,*

$$\rho_u(d_c) = 1 \quad (u \in \mathcal{L}), \qquad \rho_x(d_c) = 0 \quad (x \in C \setminus \mathcal{L}). \tag{9}$$

Proof. All cycle edges have length $d_{\mathcal{L}}$. Since distinct unordered pairs have distinct distances, the cycle must contain exactly two nodes, which are each other's unique nearest neighbors. Thus each cycle node has exactly one neighbor within distance $d_{\mathcal{L}}$.

For any $x \in C \setminus \mathcal{L}$, the cycle compactness property gives $d(x, \psi(x)) \geq d_{\mathcal{L}}$. Equality would require a different pair at the same distance as the cycle pair, contradicting the assumption. Hence $d(x, \psi(x)) > d_{\mathcal{L}}$, and x has no neighbor within the cutoff. □

4) WCC Correspondence under Graph Refinement

Sufficient Query Budget. In the first round, we add any missing final representative agents to the selected batch and process all final representative agents before the remaining selected objects. There are at most m distinct final representative agents. When inserting the j-th final representative agent, there are at most $j - 1$ existing constraint-graph WCCs. Therefore, inserting all final representative agents requires at most $\sum_{j=1}^{m}(j-1) = m(m-1)/2$ pairwise queries. The following theorem considers positive query budgets $B \geq m(m-1)/2$, a condition satisfied by all budgets used in our experiments. Under this processing order and budget condition, all final representative agents are inserted into the constraint graph by the end of the first round.

Theorem B.7 (WCC Correspondence)**.** *For every view v and round t, each weakly connected component (WCC) of the refined object graph $\mathcal{G}_t^{(v)}$ contains exactly one WCC of the constraint graph $\mathcal{G}_t^{\text{cons}}$ as a subgraph.*

Proof. We prove the theorem by contradiction, structured in two parts.
Part 1: Each refined object-graph WCC contains at most one constraint-graph WCC.

Assume for contradiction that a WCC W of $\mathcal{G}_t^{(v)}$ contains two distinct WCCs $C_{a,t}$ and $C_{b,t}$ of $\mathcal{G}_t^{\text{cons}}$, where $a \neq b$. By the insertion rule, each constraint-graph WCC starts as a singleton, and each subsequent insertion adds a new node with one outgoing ML edge to an existing node. Thus, each $C_{j,t}$, $j \in \{a, b\}$, is a directed tree with a unique root $x_{j,t}^{\text{root}}$ having out-degree zero.

In the initialized object graph, every node except the final representative agent has exactly one outgoing edge; the final representative agent has none. During refinement, outgoing edges of nodes in $\mathcal{G}_t^{\text{cons}}$ are replaced or removed, while all other nodes retain their original edges. Thus, every node in the refined object graph has at most one outgoing edge.

For each $j \in \{a, b\}$, the root $x_{j,t}^{\text{root}}$ has no outgoing edge in $\mathcal{G}_t^{\text{cons}}$. Its original outgoing edge in the object graph, if any, is therefore removed during refinement.

Let $d^+_{\mathcal{G}_t^{(v)}}(x)$ denote the out-degree of x in $\mathcal{G}_t^{(v)}$. Then

$$d^+_{\mathcal{G}_t^{(v)}}(x_{j,t}^{\text{root}}) = 0, \quad j \in \{a, b\},$$
$$d^+_{\mathcal{G}_t^{(v)}}(x) \leq 1, \quad x \in V(W).$$

Let $V(W)$ and $E(W)$ denote the node and directed edge sets of W, respectively, and let $N = |V(W)|$. Since W contains two distinct roots with out-degree zero,

$$|E(W)| = \sum_{x \in V(W)} d^+_{\mathcal{G}_t^{(v)}}(x) \leq N - 2.$$

However, W is weakly connected, so its underlying undirected graph contains a spanning tree with $N-1$ edges. Consequently, $|E(W)| \geq N - 1$, a contradiction. Therefore, W contains at most one WCC of $\mathcal{G}_t^{\text{cons}}$.
Part 2: Each refined object-graph WCC contains at least one constraint-graph WCC.

Assume for contradiction that there exists a WCC W of $\mathcal{G}_t^{(v)}$ that contains no constraint-graph WCC. Since refinement modifies outgoing edges only for nodes in the constraint graph, all nodes in W retain their original outgoing edges from $\mathcal{G}_0^{(v)}$. Thus, W is a subgraph of the initialized object graph:

$$V(W) \subseteq V(\mathcal{G}_0^{(v)}), \qquad E(W) \subseteq E(\mathcal{G}_0^{(v)}).$$

In the initialized object graph, every node except the final representative agent has exactly one outgoing edge; the final representative agent has none. By the first-round processing rule and the budget bound above, the final representative agent of view v has been inserted into the constraint graph by the end of the first round. Since inserted nodes are retained, it belongs to $\mathcal{G}_t^{\text{cons}}$ for every $t \geq 1$. Hence, it is not in W. Every node in W thus retains exactly one outgoing edge. Let $N = |V(W)|$. We obtain

$$|E(W)| = \sum_{x \in V(W)} d^+_{\mathcal{G}_t^{(v)}}(x) = N.$$

Since every node in W has exactly one outgoing edge within W, the same traversal argument as in Part 1 of the proof of Theorem B.1 implies that W contains a directed cycle.

However, $\mathcal{G}_0^{(v)}$ is weakly connected and has n nodes and $n-1$ directed edges. Its underlying undirected graph is therefore a tree. Thus, $\mathcal{G}_0^{(v)}$ has no cycles. Since W is a subgraph of $\mathcal{G}_0^{(v)}$, it also has no directed cycles, a contradiction. Therefore, every refined object-graph WCC contains at least one constraint-graph WCC.

Combining the two parts, every WCC of $\mathcal{G}_t^{(v)}$ contains exactly one WCC of $\mathcal{G}_t^{\text{cons}}$ as a subgraph, completing the proof. □

C Annotation-Noise Analysis

We evaluate the effect of imperfect pairwise feedback by independently flipping each queried must-link or cannot-link annotation with probability $\eta \in \{0, 0.01, 0.02, 0.03, 0.04\}$, where $\eta = 0$ denotes clean feedback. Stages 1–5 use the same five dataset-specific query budgets as the clustering-quality experiment, with a separate run for each budget.

Figure 1 shows that **TriGAC** generally benefits from larger query budgets at low noise rates, while noisy feedback often reduces ARI relative to clean feedback. The effect varies across datasets: on MNIST-4 and CIFAR-10, for example, some noisy curves flatten or decline at later stages. An incorrect must-link answer can place an object in a WCC containing objects from a different ground-truth cluster, and that assignment is not retracted during the run. Thus, additional queries under noisy feedback do not necessarily improve clustering quality. These results indicate that the benefit of a larger query budget depends on annotation reliability.

D Uniform Query-Budget Analysis

As a complementary analysis, we compare clustering quality under query budgets determined by dataset size. For a dataset containing n objects, we set $B = \lceil rn \rceil$, where $r \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. The same budget rule applies to all query-based methods. Each method otherwise retains its input representation and algorithm settings from the main experiment.

Tables 1 and 4 report the available results on 15 real-world datasets and one synthetic dataset. Table 2 reports Macro-ARI on the 11 small datasets for which all seven baselines have results. At each budget level, Macro-ARI averages ARI equally across these datasets; Overall averages the five budget levels. The #Cons. column reports the prescribed query budget. A3S-MV may terminate before exhausting this budget. SparseMVC and EVA-MVC do not use pairwise queries. We therefore list these three methods in the rightmost columns of the table. The ARIs of the two unsupervised methods are repeated across budget levels. A dash denotes an unavailable result. Blue cells mark a highest available ARI in a row, while green cells mark values displayed as 1.000.

Small datasets. Table 2 shows that **TriGAC** achieves the highest Macro-ARI at every uniform budget stage across the 11 small datasets. Its Macro-ARI increases from 0.536 at $0.2n$ to 0.788 at n, compared with 0.512 and 0.546 for SMCAC, the strongest baseline, at these respective stages. **TriGAC** also obtains the highest Overall Macro-ARI (0.679 versus 0.531 for SMCAC). Thus, its average advantage is present even at the smallest tested budget and becomes larger as the query budget increases.

The dataset-level results in Table 1 show that **TriGAC** achieves the highest available ARI on 5, 7, 8, 8, and 10 of the 11 datasets at the five successive budget stages. At a query budget of n, Movies is the only small dataset on which another method obtains a higher ARI.

Large and synthetic datasets. As shown in Table 4, **TriGAC** achieves the highest available ARI at all five budgets on Hdigit, at $0.8n$ and n on MNIST-10K, and from $0.4n$ onward on CIFAR-10. It does not lead on Animal at the tested budgets. Across these four large real-world datasets, **TriGAC** leads on 1, 2, 2, 3, and 3 datasets at the five successive budgets.

Paired significance tests. For each of the 11 small datasets, we average each method's ARI over the five uniform budget levels. We then compare **TriGAC** with each baseline using a two-sided paired t-test across the 11 datasets ($df = 10$), and Holm-adjust the seven p-values. Table 3 shows that **TriGAC** achieves higher average ARI than all seven baselines ($p_{\text{adj}} < 0.05$).

E Illustrative Example

Experimental setup. We construct a synthetic multi-view dataset containing 200 objects from two balanced classes. Each object has three two-dimensional views with different geometric structures: two moons, two blobs, and concentric rings. The views share the same ground-truth class labels. We compare **TriGAC**, SMCAC, PCBG, and TSMSC under pairwise constraint budgets of 10, 20, and 30. In Figs. 2–4, columns show the three original views, while rows show the ground truth followed by the four methods' results. Colors indicate cluster assignments, with predicted labels aligned to the ground truth for visualization only. Black and red segments denote directly queried must-link and cannot-link pairs, respectively; transitively inferred constraints are not shown. Each method's consensus labels are displayed across all three views, with its ARI over all 200 objects reported beside the corresponding row.

TriGAC achieves the highest ARI among the four methods at every tested budget. As the query budget increases, the partition produced by **TriGAC** more closely matches the ground-truth grouping, indicating that acquired pairwise supervision helps identify the underlying class structure. Moreover, the views exhibit different geometric structures, including nonlinear class boundaries. The results suggest that the proposed multi-view pipeline can effectively integrate view-specific structural information to support clustering under limited pairwise supervision. This illustrative example complements the quantitative evaluation on real-world datasets.

Table 1: ARI under five uniform query budgets on 11 small real-world datasets.

Dataset	Budget	#Cons.	**TriGAC**	SMCAC	PCBG	TSMSC	CSE-MV	A3S-MV	SparseMVC	EVA-MVC
3Sources	$0.2n$	34	0.628	0.562	0.504	0.553	0.267	0.152	0.361	0.350
	$0.4n$	68	0.767	0.570	0.625	0.544	0.323	0.152	0.361	0.350
	$0.6n$	102	0.932	0.559	0.641	0.558	0.367	0.252	0.361	0.350
	$0.8n$	136	1.000	0.588	0.703	0.577	0.501	0.252	0.361	0.350
	n	169	1.000	0.593	0.710	0.610	0.545	0.252	0.361	0.350
BBCSport	$0.2n$	109	0.924	0.841	0.819	0.902	0.131	0.122	0.420	0.881
	$0.4n$	218	0.982	0.824	0.836	0.934	0.170	0.122	0.420	0.881
	$0.6n$	327	1.000	0.874	0.867	0.938	0.237	0.122	0.420	0.881
	$0.8n$	436	1.000	0.897	0.879	0.958	0.307	0.122	0.420	0.881
	n	544	1.000	0.915	0.913	0.952	0.380	0.122	0.420	0.881
Movies	$0.2n$	124	0.078	0.126	-0.001	0.138	0.036	0.031	0.065	0.143
	$0.4n$	247	0.068	0.121	0.000	0.154	0.036	0.026	0.065	0.143
	$0.6n$	371	0.075	0.149	0.003	0.149	0.041	0.025	0.065	0.143
	$0.8n$	494	0.095	0.150	0.009	0.163	0.044	0.027	0.065	0.143
	n	617	0.149	0.170	0.010	0.157	0.046	0.027	0.065	0.143
WebKB	$0.2n$	211	0.925	0.918	0.005	0.003	0.304	0.023	0.585	0.086
	$0.4n$	421	0.979	0.958	0.004	0.032	0.396	0.023	0.585	0.086
	$0.6n$	631	0.992	0.962	-0.003	-0.038	0.460	0.023	0.585	0.086
	$0.8n$	841	0.992	0.971	-0.001	-0.031	0.589	0.023	0.585	0.086
	n	1051	1.000	0.979	-0.005	-0.038	0.761	0.023	0.585	0.086
Cal-5V	$0.2n$	280	0.875	0.598	0.514	0.702	0.688	0.636	0.654	0.463
	$0.4n$	560	0.963	0.678	0.670	0.702	0.732	0.636	0.654	0.463
	$0.6n$	840	0.993	0.656	0.682	0.704	0.898	0.636	0.654	0.463
	$0.8n$	1120	1.000	0.694	0.781	0.709	0.957	0.636	0.654	0.463
	n	1400	1.000	0.754	0.811	0.719	0.973	0.636	0.654	0.463
RGBD	$0.2n$	290	0.218	0.672	0.005	0.227	0.082	0.092	0.203	0.212
	$0.4n$	580	0.436	0.629	0.041	0.210	0.089	0.092	0.203	0.212
	$0.6n$	870	0.509	0.654	0.028	0.275	0.103	0.092	0.203	0.212
	$0.8n$	1160	0.594	0.762	0.048	0.269	0.117	0.092	0.203	0.212
	n	1449	0.651	0.624	0.051	0.302	0.146	0.092	0.203	0.212
YouTube	$0.2n$	400	0.131	0.119	0.000	0.082	0.128	0.173	0.198	0.052
	$0.4n$	800	0.238	0.146	0.000	0.078	0.135	0.173	0.198	0.052
	$0.6n$	1200	0.315	0.128	0.000	0.080	0.161	0.173	0.198	0.052
	$0.8n$	1600	0.383	0.188	0.000	0.085	0.189	0.173	0.198	0.052
	n	2000	0.477	0.196	0.010	0.121	0.219	0.173	0.198	0.052
Cal20	$0.2n$	478	0.680	0.548	0.321	0.319	0.550	0.489	0.535	0.120
	$0.4n$	955	0.878	0.569	0.592	0.333	0.725	0.489	0.535	0.120
	$0.6n$	1432	0.917	0.570	0.612	0.344	0.737	0.489	0.535	0.120
	$0.8n$	1909	0.955	0.611	0.642	0.397	0.755	0.489	0.535	0.120
	n	2386	0.995	0.615	0.642	0.441	0.780	0.489	0.535	0.120
Cora	$0.2n$	542	0.296	0.317	-0.006	0.433	0.019	0.004	0.016	0.311
	$0.4n$	1084	0.412	0.264	0.100	0.476	0.038	0.004	0.016	0.311
	$0.6n$	1625	0.502	0.288	0.102	0.553	0.063	0.004	0.016	0.311
	$0.8n$	2167	0.612	0.276	0.115	0.613	0.088	0.004	0.016	0.311
	n	2708	0.752	0.199	0.142	0.674	0.117	0.004	0.016	0.311
CiteSeer	$0.2n$	663	0.282	0.064	0.000	0.372	0.018	0.030	0.095	0.318
	$0.4n$	1325	0.361	0.054	0.000	0.375	0.030	0.030	0.095	0.318
	$0.6n$	1988	0.469	0.055	-0.001	0.393	0.048	0.030	0.095	0.318
	$0.8n$	2650	0.578	0.056	0.001	0.430	0.081	0.030	0.095	0.318
	n	3312	0.674	0.056	0.001	0.469	0.121	0.030	0.095	0.318
MNIST-4	$0.2n$	800	0.855	0.872	0.604	0.754	0.698	0.760	0.775	0.738
	$0.4n$	1600	0.897	0.881	0.839	0.776	0.773	0.760	0.775	0.738
	$0.6n$	2400	0.923	0.882	0.842	0.790	0.873	0.760	0.775	0.738
	$0.8n$	3200	0.945	0.898	0.871	0.810	0.932	0.760	0.775	0.738
	n	4000	0.969	0.908	0.903	0.827	0.940	0.760	0.775	0.738

Table 2: Stage-wise Macro-ARI across 11 small real-world datasets under uniform query budgets. Overall averages the five stages.

Budget	**TriGAC**	SMCAC	PCBG	TSMSC	CSE-MV	A3S-MV	SparseMVC	EVA-MVC
$0.2n$	**0.536**	0.512	0.251	0.408	0.266	0.228	0.355	0.334
$0.4n$	**0.635**	0.518	0.337	0.419	0.313	0.228	0.355	0.334
$0.6n$	**0.693**	0.525	0.343	0.431	0.363	0.237	0.355	0.334
$0.8n$	**0.741**	0.554	0.368	0.453	0.415	0.237	0.355	0.334
n	**0.788**	0.546	0.381	0.476	0.457	0.237	0.355	0.334
Overall	**0.679**	0.531	0.336	0.437	0.363	0.233	0.355	0.334

Table 3: Overall two-sided paired t-tests across 11 small datasets ($df = 10$), using ARI averaged over the five uniform query budgets. p-values are Holm-adjusted across seven comparisons.

Statistic	SMCAC	PCBG	TSMSC	CSE-MV	A3S-MV	SparseMVC	EVA-MVC
t-value	2.6978	4.5350	2.6746	4.8218	5.1565	6.2367	3.9382
p-value	4.48×10^{-2}	4.33×10^{-3}	4.48×10^{-2}	3.50×10^{-3}	2.56×10^{-3}	6.77×10^{-4}	8.35×10^{-3}

Table 4: ARI under five uniform query budgets on four large real-world datasets and one synthetic dataset. Dashes indicate unavailable results

Dataset	Budget	#Cons.	**TriGAC**	CSE-MV	A3S-MV	SparseMVC	EVA-MVC
MNIST-10K	$0.2n$	2000	0.802	0.867	0.791	0.811	0.494
	$0.4n$	4000	0.919	0.934	0.791	0.811	0.494
	$0.6n$	6000	0.952	0.979	0.791	0.811	0.494
	$0.8n$	8000	0.996	0.994	0.791	0.811	0.494
	n	10000	1.000	0.998	0.791	0.811	0.494
Hdigit	$0.2n$	2000	0.951	0.897	0.665	0.887	0.862
	$0.4n$	4000	0.987	0.903	0.665	0.887	0.862
	$0.6n$	6000	0.992	0.923	0.665	0.887	0.862
	$0.8n$	8000	0.996	0.947	0.665	0.887	0.862
	n	10000	0.999	0.993	0.665	0.887	0.862
Animal	$0.2n$	2335	0.018	0.066	0.073	0.087	0.056
	$0.4n$	4670	0.026	0.068	0.073	0.087	0.056
	$0.6n$	7004	0.047	0.071	0.073	0.087	0.056
	$0.8n$	9339	0.063	0.074	0.073	0.087	0.056
	n	11673	0.084	0.079	0.073	0.087	0.056
CIFAR-10	$0.2n$	10000	0.965	–	–	–	0.978
	$0.4n$	20000	0.986	–	–	–	0.978
	$0.6n$	30000	0.997	–	–	–	0.978
	$0.8n$	40000	0.999	–	–	–	0.978
	n	50000	1.000	–	–	–	0.978
Synthetic	$0.2n$	200000	0.980	–	–	–	0.901
	$0.4n$	400000	0.996	–	–	–	0.901
	$0.6n$	600000	0.998	–	–	–	0.901
	$0.8n$	800000	1.000	–	–	–	0.901
	n	1000000	1.000	–	–	–	0.901

References

[1] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *science*, vol. 344, no. 6191, pp. 1492–1496, 2014.

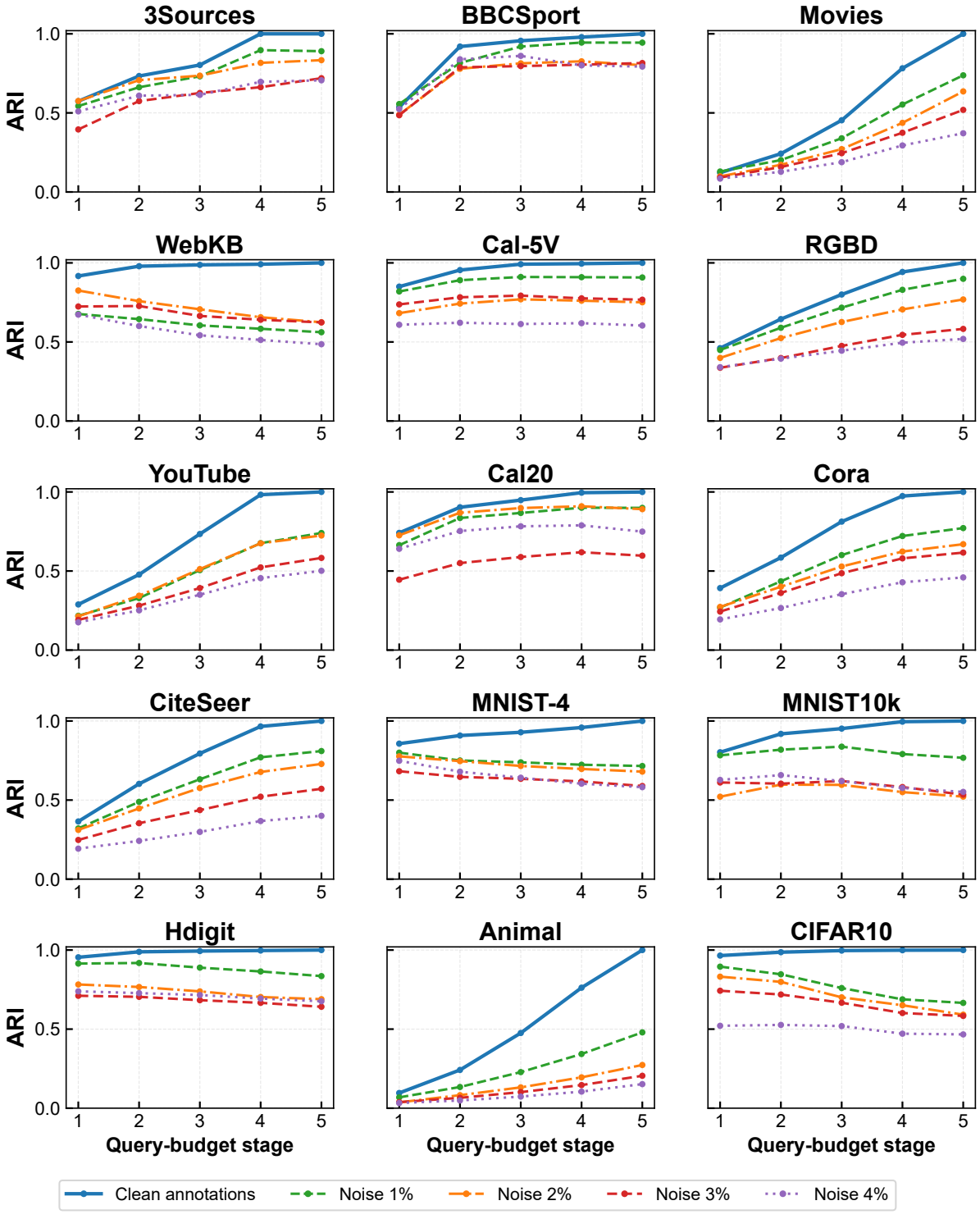


Figure 1: Stage-wise ARI on 15 real-world datasets under clean feedback and annotation-noise rates of 1%, 2%, 3%, and 4%.

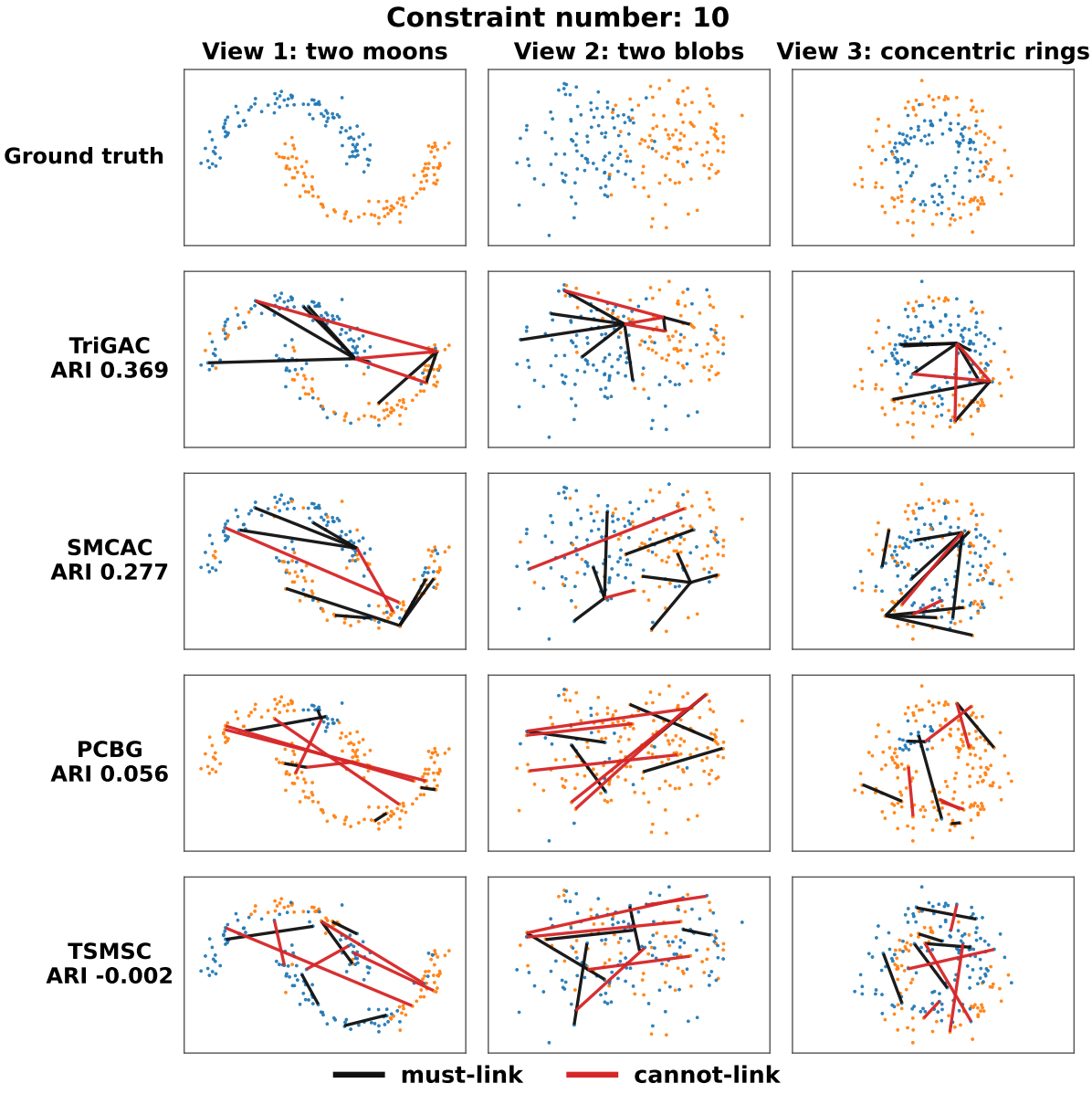


Figure 2: Clustering results with 10 pairwise constraints. Columns show the two-moons, two-blobs, and concentric-rings views. The first row shows the ground truth; the remaining rows show the results of the four methods.

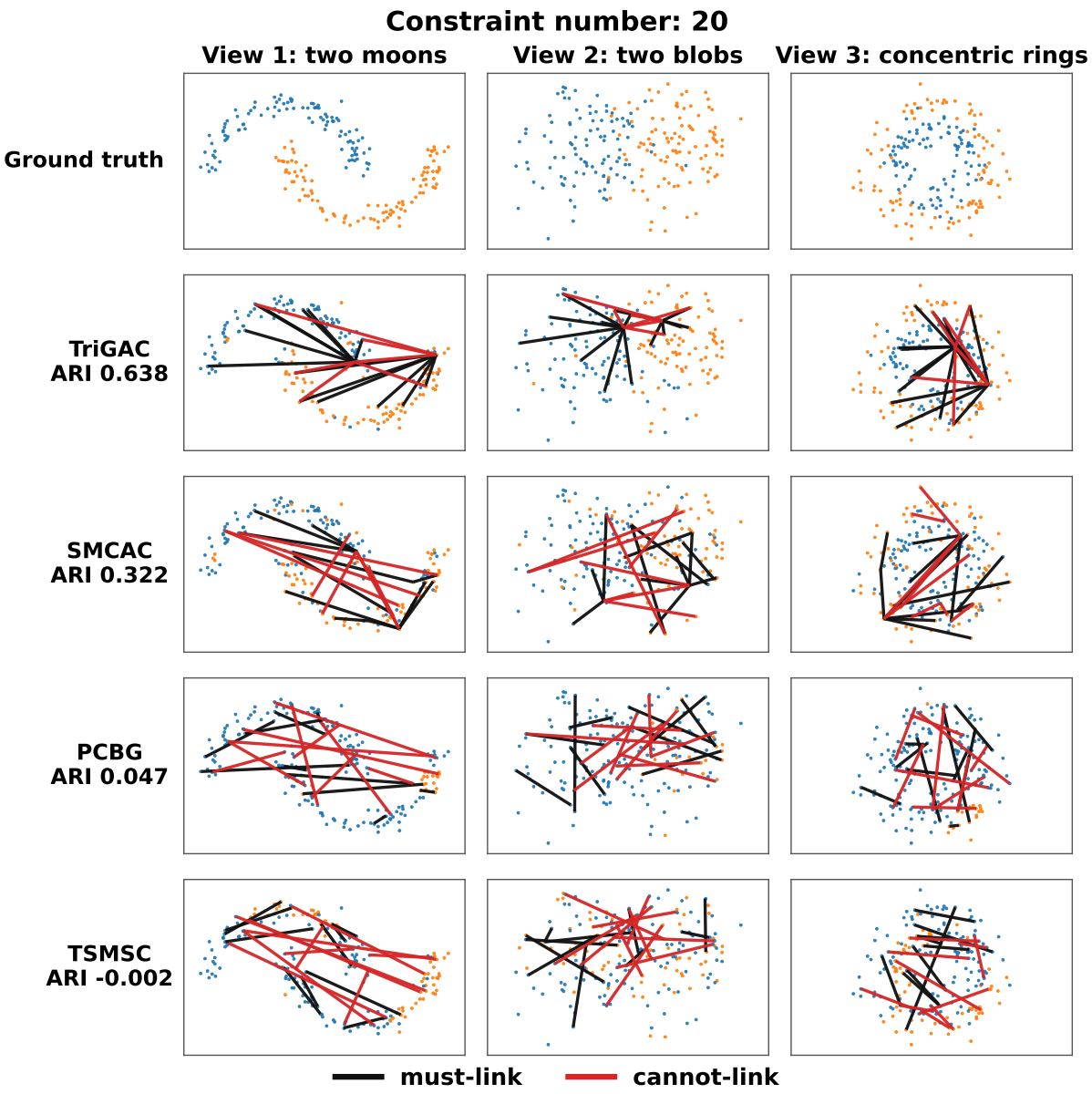


Figure 3: Clustering results with 20 pairwise constraints, using the same layout as Fig. 2.

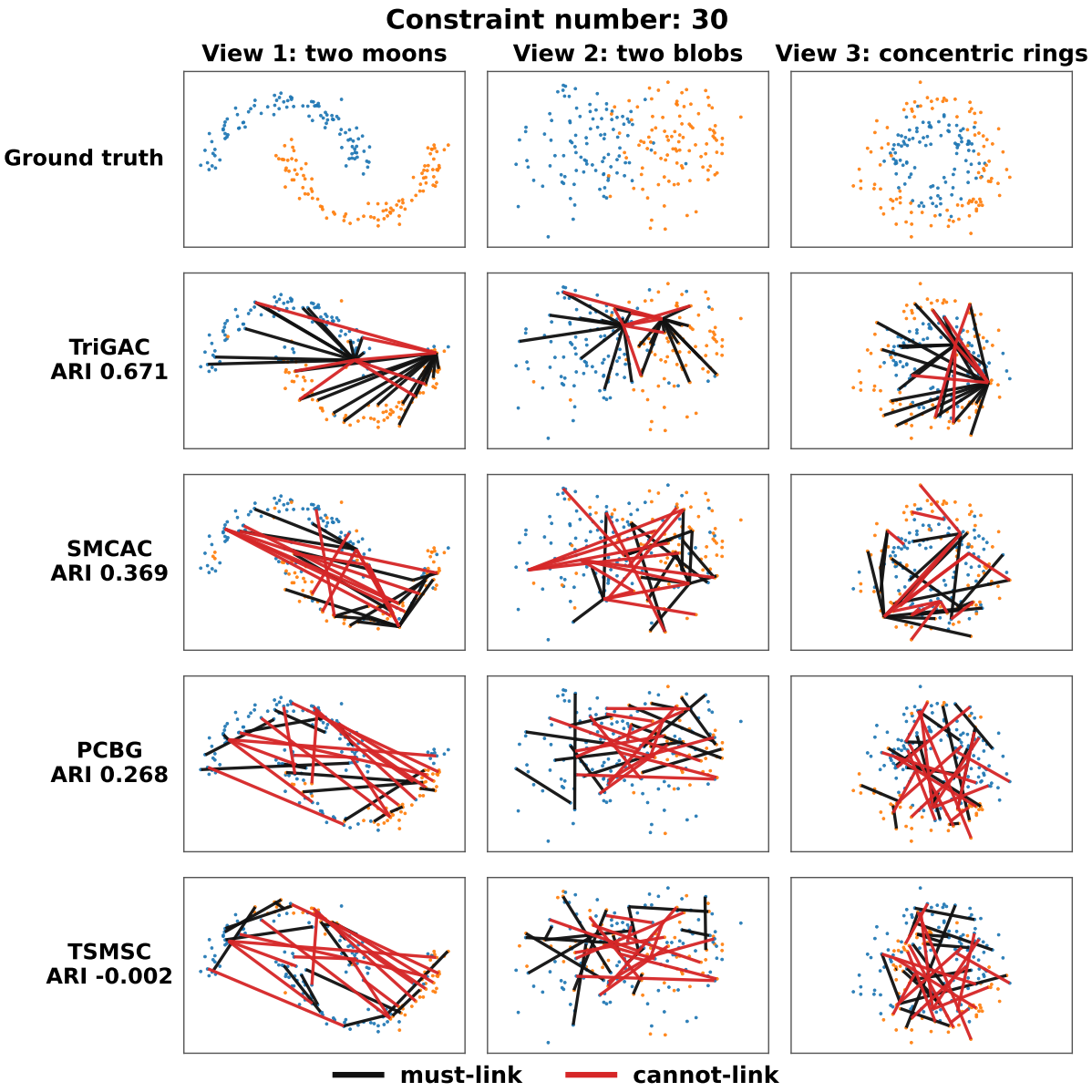


Figure 4: Clustering results with 30 pairwise constraints, using the same layout as Fig. 2.